

DATA MINING LAB

A Regional Correlation Study of Household Economics and School Enrollment in the Philippines

Analysis Results

Submitted by Group 20:

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Instructions

Try to run different data mining techniques and/or measures and analyze the results.
List down all observations here together with the actual results

Introduction

Following the instructions and data pre-processing done for the previous activity, I inserted the cleaned central dataset from the two datasets previously discussed, the Family Income and Expenditure Survey from 2017 and Philippine School Enrollment Data. Shown below is my data pre-processing flow in Altair's AI Studio.

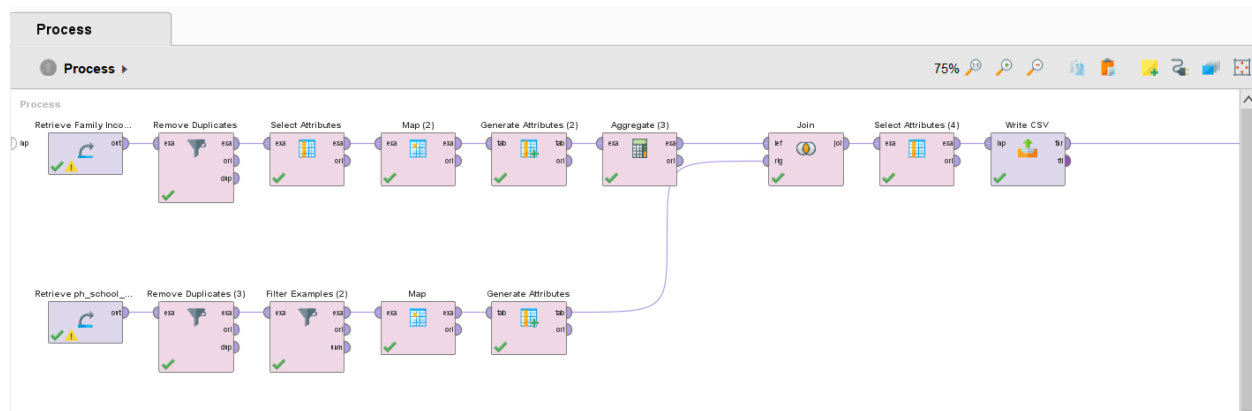


Figure 1. Data Pre-processing Workflow

Using the CSV created from this process workflow, we arrive at a table containing average annual household income for each region in the Philippines, including a breakdown of each expenditure, specifically educational expenditure, food and non-food expenditure, it also includes the number of family members, etc.. Furthermore, the combined dataset also contains the total number of enrollees for each sector (Public, Private, SUC-LUC) for each region, as well as the strands chosen for those in K-12. Following this, I have tried running multiple data mining techniques and/or measures and in this document, I will be listing each of them, explaining why I chose them and analyzing the results they gave.

Data Mining Models/Techniques/Measures Used

1. Correlation Matrix

Using correlation matrix, we will be able to identify and quantify the relationships between household socioeconomic variables and the distribution of students across various Senior High School tracks. Since the dataset includes indicators such as total household income, food and non-food expenditure, education spending, and the total number of family members, alongside counts of students enrolled in different SHS strands, the correlation matrix helps determine which socioeconomic factors are most strongly associated with track selection.

Process



Figure 2. Correlation Matrix Workflow

Results

Correlation Matrix (Correlation Matrix (2))																
Attributes	average...	average...	average...	average...	average...	average...	K10_To...	Total_S...	ABM_T...	HUMSS...	STEM...	GAS_T...	TVL_To...	MARITL...	SPORT...	A&D_T...
average(Total Household Income)	1	0.891	-0.712	0.712	0.935	-0.488	0.245	0.433	0.540	0.463	0.506	0.131	0.338	0.017	0.542	0.553
average(Education Expenditure)	0.891	1	-0.622	0.622	0.810	-0.370	0.205	0.361	0.464	0.399	0.431	0.080	0.280	-0.036	0.463	0.473
average(Food Share)	-0.712	-0.622	1	-1.000	-0.460	0.670	-0.053	-0.151	-0.267	-0.219	-0.245	0.067	-0.077	0.102	-0.252	-0.256
average(Non-Food Share)	0.712	0.622	-1.000	1	0.460	-0.670	0.053	0.151	0.267	0.219	0.245	-0.067	0.077	-0.102	0.252	0.256
average(Total Food Expenditure)	0.935	0.810	-0.460	0.460	1	-0.315	0.310	0.506	0.592	0.513	0.553	0.209	0.420	0.043	0.563	0.581
average(Total Number of Family members)	-0.488	-0.370	0.670	-0.670	-0.315	1	-0.081	-0.170	-0.165	-0.198	-0.162	-0.082	-0.159	-0.079	-0.112	-0.129
K10_Total	0.245	0.205	-0.053	0.053	0.310	-0.081	1	0.585	0.235	0.516	0.118	0.776	0.791	-0.128	0.603	0.296
Total_SHS	0.433	0.361	-0.151	0.151	0.506	-0.170	0.585	1	0.887	0.959	0.839	0.818	0.941	0.172	0.761	0.812
ABM_Total	0.540	0.464	-0.267	0.267	0.592	-0.165	0.235	0.887	1	0.898	0.982	0.476	0.691	0.220	0.674	0.871
HUMSS_Total	0.463	0.399	-0.219	0.219	0.513	-0.198	0.516	0.959	0.898	1	0.847	0.695	0.877	0.142	0.681	0.759
STEM_Total	0.506	0.431	-0.245	0.245	0.553	-0.162	0.118	0.839	0.982	0.847	1	0.412	0.610	0.320	0.584	0.832
GAS_Total	0.131	0.080	0.067	-0.067	0.209	-0.082	0.776	0.818	0.476	0.695	0.412	1	0.927	0.060	0.664	0.527
TVL_Total	0.338	0.280	-0.077	0.077	0.420	-0.159	0.791	0.941	0.691	0.877	0.610	0.927	1	0.049	0.741	0.662
MARITIME_Total	0.017	-0.036	0.102	-0.102	0.043	-0.079	-0.128	0.172	0.220	0.142	0.320	0.060	0.049	1	-0.091	0.177
SPORTS_Total	0.542	0.463	-0.252	0.252	0.563	-0.112	0.603	0.761	0.674	0.681	0.584	0.664	0.741	-0.091	1	0.838
A&D_Total	0.553	0.473	-0.256	0.256	0.581	-0.129	0.296	0.812	0.871	0.759	0.832	0.527	0.662	0.177	0.838	1

Figure 3. Correlation Matrix Results Table

Looking at the results of our correlation matrix, we can see that:

- Higher household income is moderately to strongly correlated with higher enrollment in several SHS tracks (ABM, STEM, SPORTS, A&D).
- Education expenditure shows similar positive correlations, suggesting that families who spend more on education tend to have students in specialized tracks.
- Food share has negative correlations with income-related variables, indicating that lower-income households allocate a larger portion of their budget to food.
- Tracks such as ABM, HUMSS, STEM, GAS, SPORTS, A&D are strongly correlated with each other, meaning regions with high participation in one strand often have high participation in others.
- K10 and Total SHS counts are positively correlated with almost all track totals, which is expected because they are aggregates.

2. k-Means Clustering

Using k-Means clustering, we group regions based on similarities in their socioeconomic characteristics and Senior High School (SHS) track enrollments. This method helps identify natural groupings in the dataset without any predefined labels, allowing us to see which regions share similar household patterns or SHS participation behavior.

Process

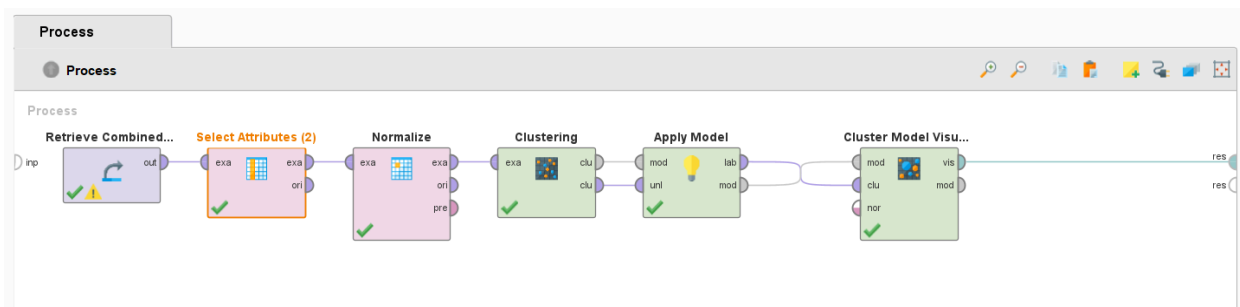


Figure 4. k-Means Clustering Workflow

Results

Number of Clusters: 3

Cluster 0

25

average(Total Number of Family members) is on average 46.69% smaller, average(Food Share) is on average 44.51% smaller, MARITIME_Total is on average 38.06% smaller

Cluster 1

21

average(Total Number of Family members) is on average 61.19% larger, average(Food Share) is on average 59.28% larger, ABM_Total is on average 48.73% smaller

Cluster 2

2

A&D_Total is on average 751.93% larger, ABM_Total is on average 727.57% larger, STEM_Total is on average 697.23% larger

Figure 5. k-Means Clustering Overview

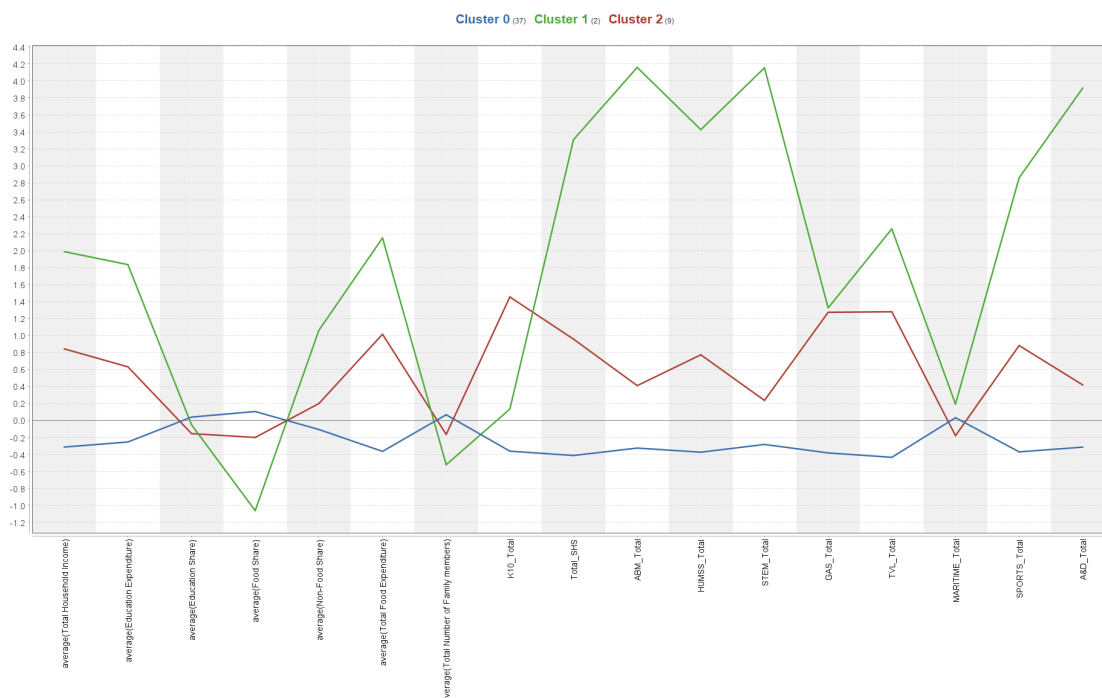


Figure 6. k-Means Clustering Line Graph

Cluster	average(...)	average(...)	average(...)	average(...)	average(...)	average(...)	average(...)	K10_Total	Total_SHS	ABM_Total	HUMSS_...	STEM_T...	GAS_Total	TVL_Total	MARITIM...	SPORTS...	A&D_Total
Cluster 0	-0.313	-0.253	0.040	0.106	-0.106	-0.364	0.068	-0.361	-0.412	-0.325	-0.373	-0.282	-0.382	-0.434	0.033	-0.369	-0.314
Cluster 1	1.989	1.837	-0.050	-1.062	1.062	2.151	-0.521	0.134	3.308	4.159	3.426	4.155	1.326	2.257	0.193	2.859	3.917
Cluster 2	0.844	0.632	-0.154	-0.199	0.199	1.017	-0.164	1.455	0.959	0.411	0.774	0.237	1.275	1.281	-0.178	0.883	0.419

Figure 7. k-Means Clustering Breakdown Table

Based on the centroid values shown in the output:

Cluster 0 (25 records)

- Smaller family size
- Lower food share
- Lower MARITIME enrollment

- Interpretation: These regions generally have smaller households and less reliance on food expenditure, reflected in lower participation in the Maritime strand.

Cluster 1 (21 records)

- Larger family size
- Higher food share
- Much lower ABM enrollment
- Interpretation: These regions represent larger, likely lower-income households (high food share). They also show significantly low ABM participation.

Cluster 2 (2 records)

- Extremely high A&D, ABM, and STEM enrollments
- Interpretation: This small cluster consists of highly specialized or urbanized regions where academic and creative SHS tracks are unusually popular.

3. Decision Tree Classification

The decision tree model is used to determine which Senior High School (SHS) strand is most likely to be chosen based on household socioeconomic characteristics. By examining how variables such as food share, education spending, household income, age structure, and school sector relate to track selection, the model identifies the most important predictors that explain why a region or group tends to choose GAS, TVL, STEM, or ABM.

Process

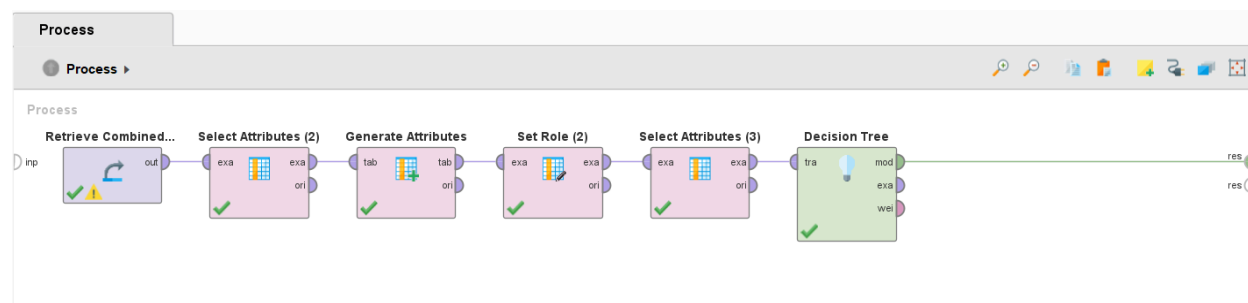


Figure 8. Decision Tree Classification Workflow

Results

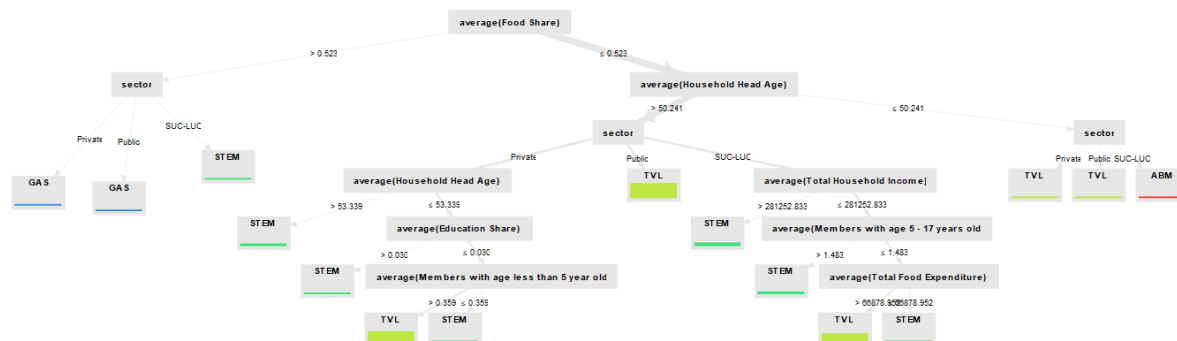


Figure 9. Decision Tree Diagram

The decision tree shows that Food Share is the strongest predictor of strand choice. Regions with high food share are often an indicator of lower income, and are more likely to select GAS, while regions with lower food share show more diverse outcomes.

Among low-food share regions, Household Head Age becomes the next major factor. Areas with older household heads tend to choose STEM or ABM, while those with younger household heads, especially in public-school areas, frequently choose TVL.

School Sector also plays a significant role:

- Public schools lean toward TVL,
- Private and SUC-LUC schools favor STEM or ABM,
- High-food share in private regions still tends toward GAS.

Other socioeconomic variables such as income, education spending, and household age composition help refine the final classification. Higher income and higher education spending typically push regions toward STEM, while lower income tends to direct them toward TVL.

Overall, the tree indicates:

1. GAS - high food share (lower-income households)
2. TVL - public-sector, younger household heads, lower income
3. STEM - older household heads, higher income, higher education spending
4. ABM - low food share, higher income, private/SUC-LUC sectors